The documentation and process conversion measures necessary to comply with this revision shall be completed by 15 December 1993

INCH-POUND

MIL-S-19500/375C
15 September 1993
SUPERSEDING
MIL-S-19500/375B
16 August 1982

MILITARY SPECIFICATION
SEMICONDUCTOR DEVICE, TRANSISTOR, FIELD-EFFECT,
N-CHANNEL, DEPLETION MODE, SILICON
TYPES 2N3821, 2N3822, 2N3823
JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for N-channel, junction, silicon, field-effect depletion mode transistors. Three levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See 3.3.

1.3 Maximum ratings.

P_T 1/	V_{DG} and V_{DS}			V_{GSR}			I_{GF}	T_{op} and T_{STG}
T_A = +25°C	2N3821	2N3822	2N3823	2N3821	2N3822	2N3823		
mW	V dc	V dc	V dc	V dc	V dc	V dc	mA dc	°C
300	50	50	30	50	50	30	10	-55 to +200

1/ Derate linearly, 1.7 mW/°C for $T_A \geq$ +25°C.

1.4 Primary electrical characteristics.

	I_{DSS} 1/			$V_{GS(off)}$			NF			
	V_{DS} = 15 V dc, V_{GS} = 0			V_{DS} = 15 V dc, I_D = 0.5 nA dc			V_{DS} = 15 V dc, V_{GS} = 0, R_G = 1 MΩ, f = 10 Hz		V_{DS} = 15 V dc, V_{GS} = 0, R_G = 1 MΩ, f = 1 kHz	V_{DS} = 15 V dc, V_{GS} = 0, R_G = 1 kΩ, f = 105 MHz
	2N3821	2N3822	2N3823	2N3821	2N3822	2N3823	2N3821	2N3822	All types	2N3823
	mA dc	mA dc	mA dc	V dc	V dc	V dc	dB		dB	dB
Min	0.5	2.0	4.0							
Max	2.5	10.0	20.0	4.0	6.0	8.0	5.0		2.5	2.5

1/ Pulsed (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: NASA/Parts Projects Office (NPPO), NASA Goddard Space Flight Center, Code 310.A, Greenbelt, MD 20771 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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	$\lvert y_{fs} \rvert$ 1 1/ V_{DS} = 15 V dc, f = 1 kHz, V_{GS} = 0			$\lvert y_{fs} \rvert$ 2 V_{DS} = 15 V dc, V_{GS} = 0			C_{iss} V_{DS} = 15 V dc, V_{GS} = 0, 100 kHz ≤ f ≤ 1 MHz	C_{rss} V_{DS} = 15 V dc, V_{GS} = 0, 100 kHz ≤ f ≤ 1 MHz		
				f = 100 MHz	f = 100 MHz	f = 200 MHz				
	2N3821	2N3822	2N3823	2N3821	2N3822	2N3823		2N3821	2N3822	2N3823
	µS	µS	µS	µS	µS	µS	pF	pF	pF	pF
Min	1,500	3,000	3,500	1,500	3,000	3,200				
Max	4,500	6,500	6,500				6	3	3	2

1/ Pulsed; see 4.5.1.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

R_{ISO} - - - - - Insulation resistance, gate lead to case.

V_1 - - - - - - rms value of the specified reference voltage in the specified circuit.

V_2 - - - - - - rms value of the measured voltage between the specified points in the specified circuit.

$\lvert y_{os} \rvert$ - - - - Magnitude of small-signal common-source short-circuit output admittance.

S - - - - - - Unit of conductance or admittance (formerly mho).

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3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, appendix F, figure 10.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-S-19500. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
11	I_{GSSR1}, I_{DSS}, and $\|y_{fs}\|1$	I_{GSSR1}, I_{DSS}, and $\|y_{fs}\|1$
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; ΔI_{DSS} = ±10 percent of initial value; $\Delta\|y_{fs}\|1$ = ±20 percent initial value	Subgroup 2 of table I herein; ΔI_{DSS} = ±10 percent of initial value; $\Delta\|y_{fs}\|1$ = ±20 percent of initial value.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

2N3821 and 2N3822 = V_{GS} = 40 V dc; V_{DS} = 0; T_A = +150°C

2N3823 = V_{GS} = 24 V dc; V_{DS} = 0; T_A = +150°C.

NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JANTX) of MIL-S-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

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4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

Subgroup	Method	Condition
B4	1037	2N3821 and 2N3822: V_{DS} = 30 V dc, V_{GS} = 0; 2N3823: V_{DS} = 15 V dc, V_{GS} = 0; at T_A = +30°C ±5°C. t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. No heat sink or forced-air cooling on the devices shall be permitted.
B5	1027	V_{DS} = 0; 2N3821 and 2N3822: V_{GS} = 40 V dc; 2N3823: V_{GS} = 24 V dc; at T_A = +225°C for 168 hours or T_A = +200°C for 260 hours.

4.4.2.2 Group B inspection, table IVb (JANTX and JANTXV) of MIL-S-19500.

Subgroup	Method	Condition
B3	1027	2N3821 and 2N3822: V_{GS} = 40 V dc, V_{DS} = 0; 2N3823: V_{DS} = 0, V_{GS} = 24; at T_A = +150°C ±5°C.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

Subgroup	Method	Condition
C2	2036	Test condition E.
C6	1026	2N3821 and 2N3822: V_{GS} = 40 V dc, V_{DS} = 0; 2N3823: V_{DS} = 0, V_{GS} = 24; at T_A = +150°C ±5°C.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Disposition of case lead during electrical measurements. All electrical measurements (except for case insulation test) operating life test shall be performed with the case lead connected to the source.

4.5.2.1 Case insulation resistance test. Case insulation resistance test shall be measured between the case and gate lead.

4.5.3 Small-signal common-source short-circuit input (or output) conductance. These tests shall be conducted with a General Radio transfer function and admittance bridge model 1607A (or suitable equivalent) in accordance with the portion of its accompanying handbook which is applicable to this measurement. A Hewlett-Packard generator model 608D and a Nems-Clarke receiver model 1502A (or suitable equivalents) shall be used with the transfer function and admittance bridge.

4.5.4 10 Hz and 1 kHz spot noise figure tests. These tests shall be conducted with a model 2173C Quan Tech Laboratories test set, or equivalent. Conditions shall be as specified in table I.

4.5.5 105 MHz spot noise figure test. This test shall be conducted with the equipment and circuit shown on figures 1 and 2, or suitable equivalent.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Gate - source breakdown voltage	3401	Bias condition C; I_G = 1.0 µA dc; V_{DS} = 0	$V_{(BR)GSSR}$			
2N3821, 2N3822				50		V dc
2N3823				30		V dc
Gate reverse current	3411	Bias condition C;	I_{GSSR1}			
2N3821, 2N3822		V_{GS} = 30 V dc; V_{DS} = 0			0.1	nA dc
2N3823		V_{GS} = 20 V dc; V_{DS} = 0			0.5	nA dc
Zero-gate-voltage drain current	3413	Bias condition C; V_{DS} = 15 V dc; V_{GS} = 0; pulsed (see 4.5.1)	I_{DSS}			
2N3821				0.5	2.5	mA dc
2N3822				2.0	10.0	mA dc
2N3823				4.0	20.0	mA dc
Gate - source voltage	3403	V_{DS} = 15 V dc;	V_{GS}			
2N3821		I_D = 50 µA dc		0.5	2.0	V dc
2N3822		I_D = 200 µA dc		1.0	4.0	V dc
2N3823		I_D = 400 µA dc		1.0	7.5	V dc
Gate - source cutoff voltage	3403	V_{DS} = 15 V dc; I_D = 0.5 nA dc	$V_{GS(off)}$			
2N3821					4.0	V dc
2N3822					6.0	V dc
2N3823					8.0	V dc
Case insulation resistance	1016	Test condition A; (see 4.5.2.1)	R_{ISO}	10^{10}		ohm
Subgroup 3						
High-temperature operation:		T_A = +150°C				
Gate reverse current	3411	Bias condition C;	I_{GSSR2}			
2N3821, 2N3822		V_{GS} = 30 V dc; V_{DS} = 0			0.1	µA dc
2N3823		V_{GS} = 20 V dc; V_{DS} = 0			0.5	µA dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 3 - Continued						
Low-temperature operation:		T_A = -55°C				
Small-signal, common-source, short-circuit, forward transadmittance	3455	V_{DS} = 15 V dc; V_{GS} = 0; f = 1 kHz; pulsed (see 4.5.1 and 4.4.7)	$\|y_{fs}\|1$			
2N3821					6,750	µS
2N3822					9,750	µS
2N3823					9,750	µS
Subgroup 4						
Small-signal common-source, short-circuit, forward transadmittance	3455	V_{DS} = 15 V dc; V_{GS} = 0; f = 1 kHz; pulsed (see 4.5.1)	$\|y_{fs}\|2$			
2N3821				1,500	4,500	µS
2N3822				3,000	6,500	µS
2N3823				3,500	6,500	µS
Small-signal common-source, short-circuit, output admittance	3453	V_{DS} = 15 V dc; V_{GS} = 0; f = 1 kHz; pulsed (see 4.5.1)	$\|y_{os}\|$			
2N3821					10	µS
2N3822					20	µS
2N3823					35	µS
Small-signal common-source, short-circuit, input capacitance	3431	V_{DS} = 15 V dc; V_{GS} = 0; 100 kHz ≤ f ≤ 1 MHz	C_{iss}		6	pF
Small-signal common-source, short-circuit, reverse-transfer capacitance	3453	V_{DS} = 15 V dc; V_{GS} = 0; 100 kHz ≤ f ≤ 1 MHz	C_{rss}			
2N3821, 2N3822					3	pF
2N3823					2	pF
Small-signal common-source, short-circuit, forward transadmittance	3455	V_{DS} = 15 V dc; V_{GS} = 0	$\|y_{fs}\|3$			
2N3821		f = 100 MHz		1,500		µS
2N3822		f = 100 MHz		3,000		µS
2N3823		f = 200 MHz		3,200		µS

See footnote at end of table.

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NOTES:

TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued						
Small-signal common-source, short-circuit, input conductance		V_{DS} = 15 V dc; V_{GS} = 0; f = 200 MHz; (see 4.5.3)	g_{is}			
2N3823 only					800	µS
Small-signal common-source, short-circuit, output conductance		V_{DS} = 15 V dc; V_{GS} = 0; f = 200 MHz; (see 4.5.3)	g_{os}			
2N3823 only					200	µS
Common-source spot noise figure		V_{DS} = 15 V dc; V_{GS} = 0; R_G = 1 MΩ; (see 4.5.4)	NF_1			
2N3821, 2N3822 only		f = 10 Hz			5.0	dB
All three types		f = 1 kHz			2.5	dB
Common-source spot noise figure		V_{DS} = 15 V dc; V_{GS} = 0; R_G = 1 kΩ; f = 105 MHz; (see 4.5.5)	NF_2			
2N3823 only					2.5	dB
Subgroups 5, 6, and 7						
Not applicable						

1/ For sampling plan, see MIL-S-19500.

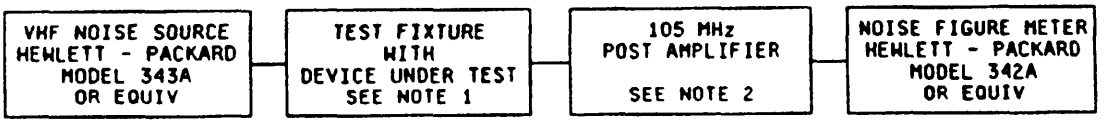
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TABLE II. Group B and C electrical measurements. 1/ 2/ 3/

Step	Inspection	MIL-STD-750		Symbol	Limit		Unit
		Method	Conditions		Min	Max	
1.	Gate reverse current	3411	Bias condition C	I_{GSSR1}			
	2N3821, 2N3822		V_{GS} = 30 V dc; V_{DS} = 0			0.1	nA dc
	2N3823		V_{GS} = 20 V dc; V_{DS} = 0			0.5	nA dc
2.	Gate reverse current	3411	Bias condition C	I_{GSSR3}			
	2N3821, 2N3822		V_{GS} = 30 V dc; V_{DS} = 0			1.0	nA dc
	2N3823		V_{GS} = 20 V dc; V_{DS} = 0			5.0	nA dc
3.	Zero-gate voltage drain current	3413	Bias condition C; V_{DS} = 15 V dc; V_{GS} = 0 Pulsed (see 4.5.1)	I_{DSS}			
	2N3821				0.5	2.5	mA dc
	2N3822				2.0	10.0	mA dc
	2N3823				4.0	20.0	mA dc
4.	Small-signal common-source, short-circuit forward transadmittance	3455	V_{DS} = 15 V dc; V_{GS} = 0; f = 1 kHz; Pulsed (see 4.5.1)	$\|y_{fs}\|1$			
	2N3821				1,500	4,500	µS
	2N3822				3,000	6,500	µS
	2N3823				3,500	6,500	µS
5.	Gate - source voltage	3403	V_{DS} = 15 V dc	ΔV_{GS}		±50 mV change from previously measured value.	
	2N3821		I_D = 50 µA dc				
	2N3822		I_D = 200 µA dc				
	2N3823		I_D = 400 µA dc				
6.	Zero-gate-voltage drain current	3413	Bias condition C; V_{DS} = 15 V dc; V_{GS} = 0; pulsed (see 4.5.1)	ΔI_{DSS}		±15 percent change from previously measured value.	
7.	Small-signal, common-source short-circuit, forward transadmittance	3455	V_{DS} = 15 V dc; V_{GS} = 0; f = 1 kHz; pulsed (see 4.5.1)	$\Delta\|y_{fs}\|1$		±25 percent change from previously measured value.	

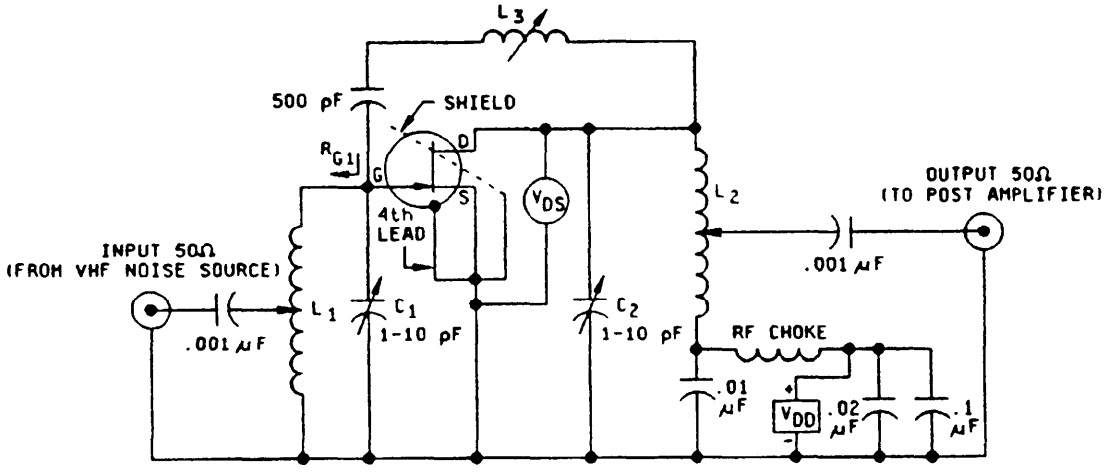
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NOTES:
1. The test fixture and procedure shall be in accordance with figure 2.
2. The post amplifier shall contain suitable biasing circuits and shall be constructed utilizing very high frequency design technique.

FIGURE 1. Block diagram for 105 MHz spot noise figure test.



CIRCUIT-COMPONENT INFORMATION:

L_1: Approximately 8.5 turns of No. 16 tinned copper; diameter approximately .375"; length approximately .9"; tapped at approximately 2.5 turns (adjust to give R_{G1} = 100 Ω); parallel resistance = 40 kΩ; tunes at approximately 8 pF.

L_2: Approximately 13.5 turns of No. 16 tinned copper; diameter approximately .375"; length approximately 1.2"; tapped at approximately 5 turns; parallel resistance = 40 kΩ; tunes at approximately 4 pF.

L_3: Approximately 15 turns of No. 20 tinned copper.

TEST PROCEDURE:

1. Neutralization shall be accomplished as follows:
   a. Insert a typical device from the units to be tested.
   b. Using a sweep generator and an oscilloscope adjust the input (C_1) and output (C_2) for maximum gain at 105 MHz.
   c. Reverse the input and output of the test jig and tune L_3 for a null.
   d. Return the input and output of the test jig to normal.
2. Once neutralization has been obtained, the devices may be tested by inserting them in the test jig and adjusting C_1 and C_2 to obtain minimum noise figure for each device.

FIGURE 2. 105 MHz spot noise figure test fixture.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish (see 3.3.1).

c. Type designation and product assurance level.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, AV, MI, SM
Navy - AS, CG, MC
Air Force - 13, 15, 19, 85, 99
DLA - ES

Preparing activity:
NASA - NA

Agent:
DLA - ES

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/375C	2. DOCUMENT DATE (YYMMDD) 93/09/15

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, TRANSISTOR, FIELD-EFFECT, N-CHANNEL, DEPLETION MODE, SILICON
TYPES 2N3821, 2N3822, 2N3823, JANTX, JANTXV, AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON 301-731-8694
c. ADDRESS (Include Zip Code) NASA/Parts Projects Office (NPPO) NASA Goddard Space Flight Center Code 310.A Greenbelt, MD 20771	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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